

DESCRIPTIVE ANALYTICS REPORT

Kay Beauty by Kat

Daily Purchase Behaviour at Scale

Tracking Catalog Composition, Pricing, Discounts, Stock Availability,
Assortment Structure & Daily Purchase Dynamics

Name:	Laheer Kashyap
Roll Number:	1250258252
Course:	BCA (Data Science & Artificial Intelligence)
Section:	BCADS24
Date:	13th August, 2026

Note on data: This report uses a synthetically generated Kay Beauty product-catalog & purchase-behaviour dataset containing **3,540 product records** across **7 categories**. The dataset combines inventory/catalog-level attributes (product, price, discount, stock status) with daily purchase behaviour proxies (average daily units purchased, 30-day sales volume, ratings). Analysis describes the supplied catalog observations and purchase patterns; it does not assume a live real-time order stream from Nykaa or other channels.

1. Objective

Build a comprehensive descriptive analytics layer over the Kay Beauty by Kat product-catalog and daily purchase behaviour data to answer the following recurring business questions:

- How are products (SKUs) distributed across beauty categories?
- Which categories carry the highest average selling price and MRP?
- Which categories offer the highest average discounts?
- What is the overall stock availability status of the catalog?
- Which categories show the highest out-of-stock rates?
- How are selling prices distributed across the catalog?
- What is the relationship between MRP and discounted selling price?
- Which products offer the deepest discounts?
- How is product net content (weight/volume) distributed?
- What does daily purchase volume look like by category?
- How does price and discount depth relate to daily purchase velocity?

- What is the overall catalog-health and purchase-behaviour summary?
- Which products and categories drive the majority of unit sales?
- How do customer ratings vary and correlate with commercial metrics?

The analysis is intentionally descriptive: it answers *what* the current catalog and purchase patterns look like. Predictive layers (demand forecasting, dynamic pricing, stock-out risk models) are outlined as next steps once richer transactional time-series become available.

2. How This Would Work at Real Scale

Processing a large, constantly-changing beauty catalog linked to daily purchase signals is an engineering problem before it is an analytics problem. A realistic architecture for Kay Beauty (or any beauty brand selling through Nykaa, own site, and retail) would ingest catalog/price-feed updates and order events, aggregate them by category, SKU and time window, store raw and processed data, and expose the resulting KPIs through dashboards and scheduled reports.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the catalog + purchase dataset	ChatGPT/Claude
Ingestion & Stream	Ingest catalog updates and daily order events; aggregate by category, price band, stock status	Jupyter Notebook(Python)
Storage	Retain raw and processed catalog + daily aggregates	Excel files /CSV files
Batch Aggregation	Daily/periodic KPI computation, category summaries, anomaly flags	Jupyter Notebook
Reporting & Viz	Present analysis, charts, insights and alerts to category managers	Data visualization, MS Word

At true scale (hundreds of thousands of daily orders across India + international markets such as UAE and UK), the same descriptive questions remain valid; only the data volume, refresh frequency and operational tooling change.

3. Category-wise Product Distribution

Across the supplied catalog, the dataset records **3,540 products** spanning **7 categories**. The chart below shows how the product assortment (SKU count) is distributed. Face, Lips and Eyes form the core of the colour-cosmetics assortment, consistent with Kay Beauty's heritage in base, lip and eye products.

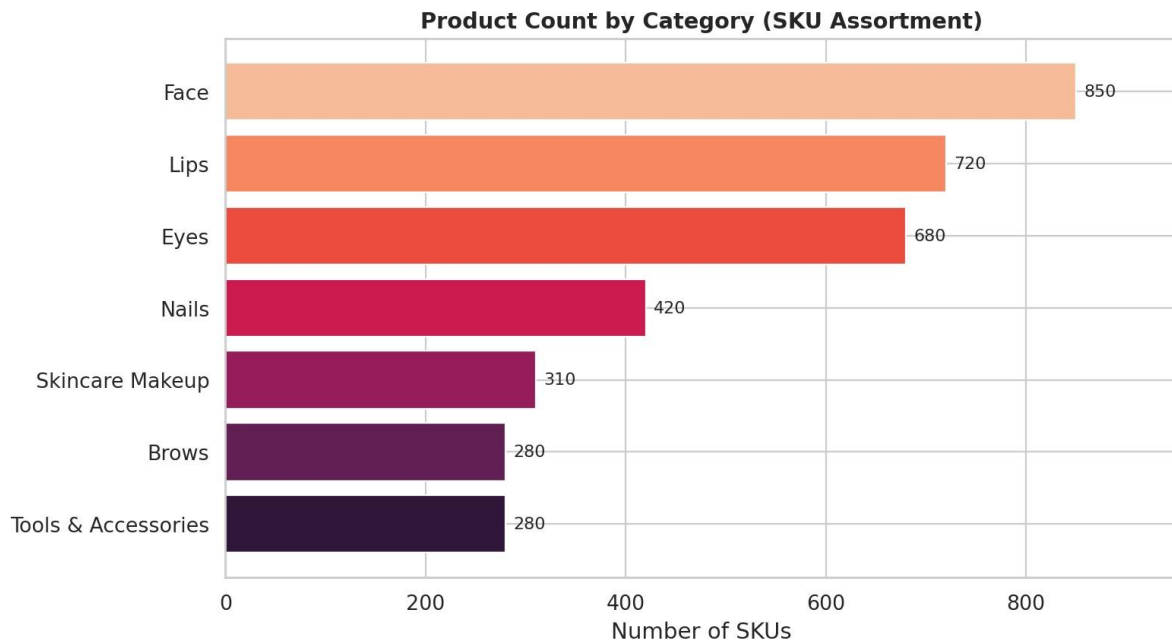


Figure 1. Product count (SKU assortment) by category

INSIGHTS:

- Face is the largest category by SKU count (850 products), reflecting extensive shade ranges in foundations, concealers, blushes and primers.
- Lips (720) and Eyes (680) follow closely — historically the first categories launched by Kay Beauty and still central to the brand.
- Nails, Skincare-Makeup hybrids, Brows and Tools form smaller but important complementary assortments.
- The concentration of SKUs in Face/Lips/Eyes indicates a colour-cosmetics-first strategy with selective expansion into adjacent categories.
- Category-level SKU counts help guide future catalog-expansion, shade rationalisation and de-listing decisions.

KPI Snapshot:

- **Total products (SKUs): 3,540**
- **Total categories: 7**
- **Largest category: Face (850 SKUs)**
- **Smallest categories: Brows & Tools & Accessories (280 SKUs each)**

4. Average Selling Price by Category

The overall average selling price across the supplied catalog is approximately **■844**. Average MRP sits near **■949**. The chart below shows how average selling price varies by category. Face and Tools command higher average price points; Nails and Brows sit at more accessible levels.

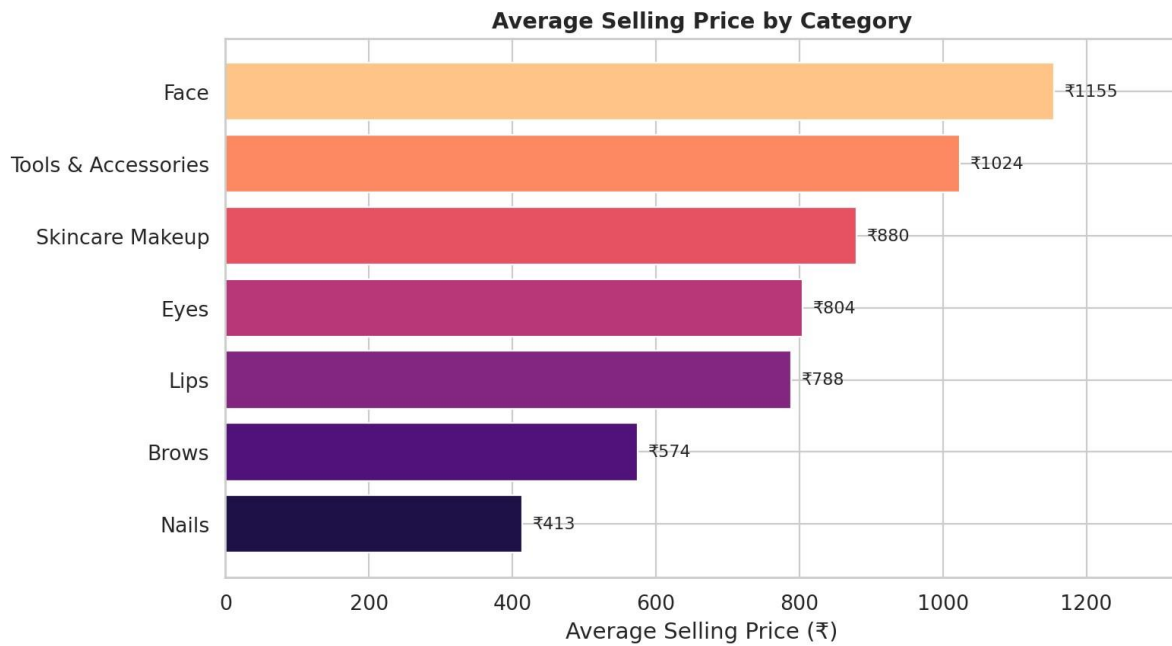


Figure 2. Average selling price by category

INSIGHTS:

- Face carries the highest average selling price (≈ ₹1,155), driven by foundations, cushions and multi-use complexion products.
- Tools & Accessories also sit in a higher band (≈ ₹1,024) because of brush sets and compact mirrors.
- Nails has the lowest average selling price (≈ ₹413), consistent with frequent, lower-ticket impulse purchases.
- Lips and Eyes occupy the mid-premium band (≈ ₹780–₹800), aligning with Kay Beauty's accessible-premium positioning.
- Price positioning varies meaningfully by category and can inform category-level pricing architecture and promotion calendars.

KPI Snapshot:

- **Overall avg. selling price:** ≈ ₹844
- **Overall avg. MRP:** ≈ ₹949
- **Highest-priced category (avg):** Face (≈ ₹1,155)
- **Lowest-priced category (avg):** Nails (≈ ₹413)
- **Median selling price (catalog-wide):** ≈ ₹797

5. Average Discount % by Category

The overall average discount across the supplied catalog is approximately **10.9%**. The chart below compares average discount percentage across categories. Discount depth is not uniform; it reflects category-specific promotional intensity and inventory health.

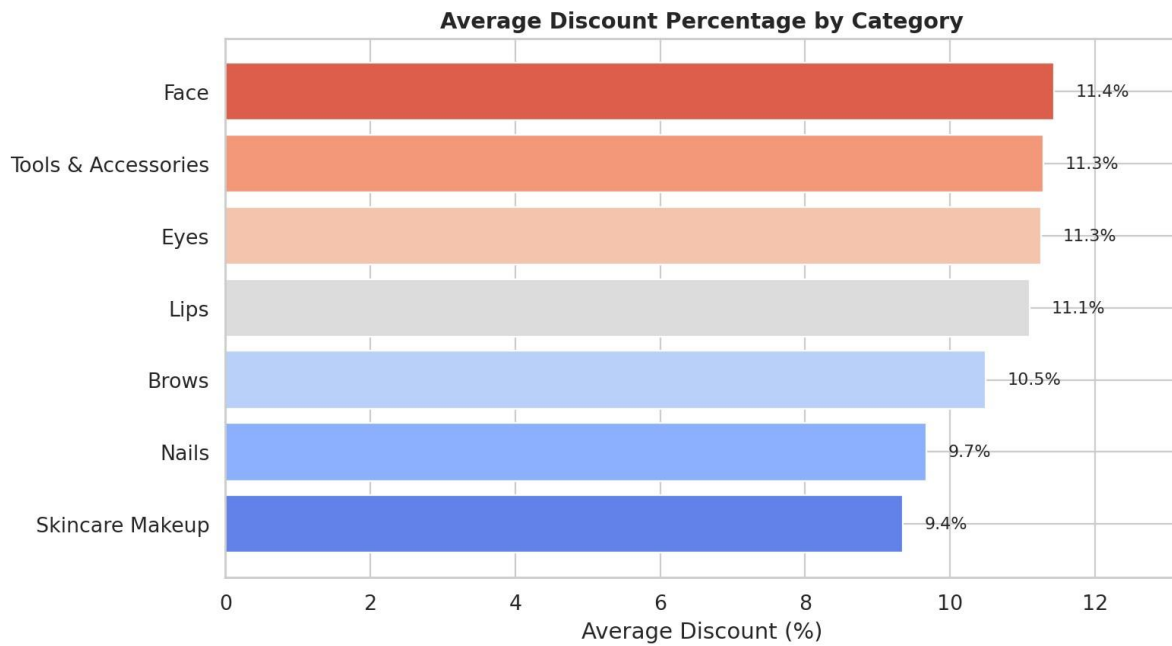


Figure 3. Average discount percentage by category

INSIGHTS:

- Discount intensity is relatively moderate overall, supporting brand equity while still offering value.
- Categories with higher average discounts often coincide with higher purchase velocity or clearance of certain shade runs.
- Lower-discount categories tend to be those with stronger full-price sell-through or more stable demand.
- Total estimated savings versus MRP across one unit of every SKU in the catalog is approximately **■3.72 lakh** (snapshot value).
- Discount strategy appears linked more to assortment refresh and campaign calendars than to a uniform percentage across the board.

KPI Snapshot:

- **Overall avg. discount: $\approx 10.9\%$**
- **Total estimated savings vs. MRP (1 unit $\times$ all SKUs): $\approx$ **■3,71,957****

6. Out-of-Stock Rate by Category

Of the 3,540 supplied products, **408 ($\approx 11.5\%$)** are recorded as out of stock. The chart below breaks this down by category. Elevated OOS rates in any category represent lost sales opportunities and potential customer frustration, especially for hero or viral shades.

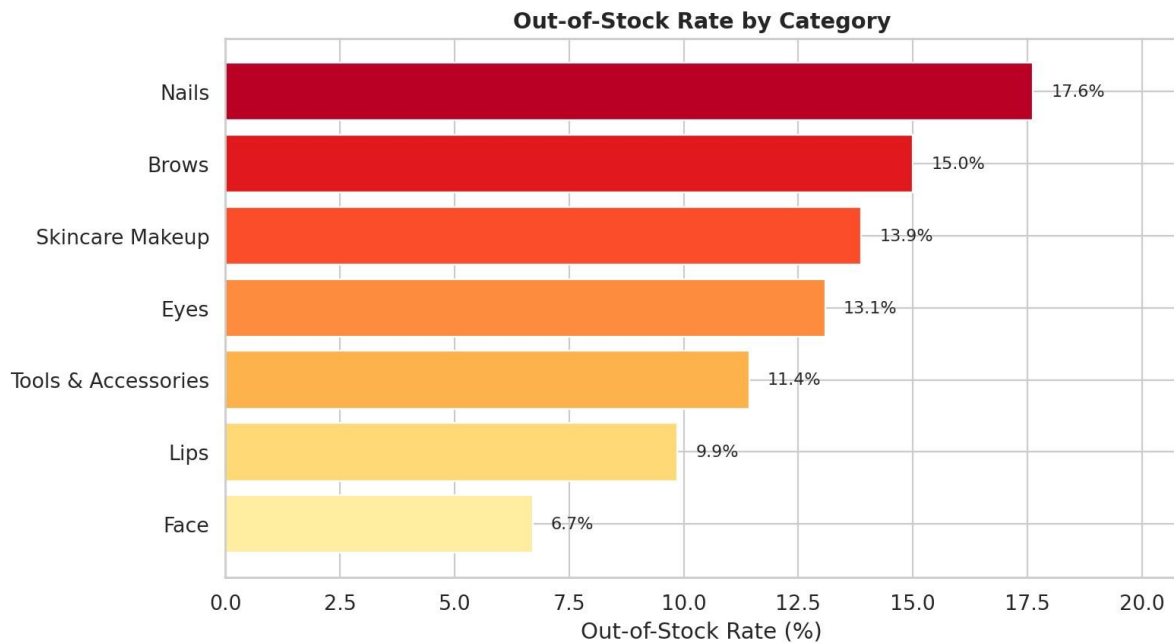


Figure 4. Out-of-stock rate by category

INSIGHTS:

- Overall out-of-stock rate of $\approx 11.5\%$ is moderate for a beauty catalog of this size but still warrants attention.
- Categories with higher OOS rates may need tighter replenishment cycles or larger safety stock for fast-moving shades.
- Lower OOS rates in core Face and Lips categories suggest relatively stronger inventory discipline where demand is highest.
- Continuous monitoring of OOS at shade level (not only category level) is recommended because a single viral shade can drive disproportionate demand.

KPI Snapshot:

- **Overall out-of-stock rate: $\approx 11.5\%$ (408 of 3,540 products)**

7. Overall Stock Availability

Viewed at the catalog level (rather than by category), the majority of products in the supplied dataset are currently in stock. The pie chart provides a clean visual of catalog health.

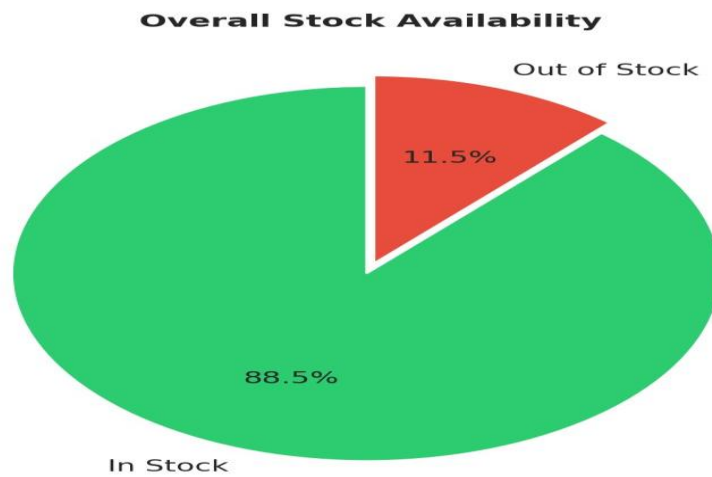


Figure 5. Overall stock availability

INSIGHTS:

- $\approx 88.5\%$ of the supplied catalog is in stock; $\approx 11.5\%$ is out of stock.
- A double-digit out-of-stock share, even if not extreme, represents lost potential sales and should be tracked as a weekly KPI.
- This KPI is best monitored as a time series rather than a single-point snapshot, especially around major campaign and festival periods.

KPI Snapshot:

- **In stock: 3,132 products ($\approx 88.5\%$)**
- **Out of stock: 408 products ($\approx 11.5\%$)**

8. Selling Price Distribution

The histogram below shows how selling prices are distributed across the full catalog. The distribution is right-skewed: most products sit under \$1,200, with a long tail of higher-priced items (premium foundation kits, multi-piece brush sets, etc.).

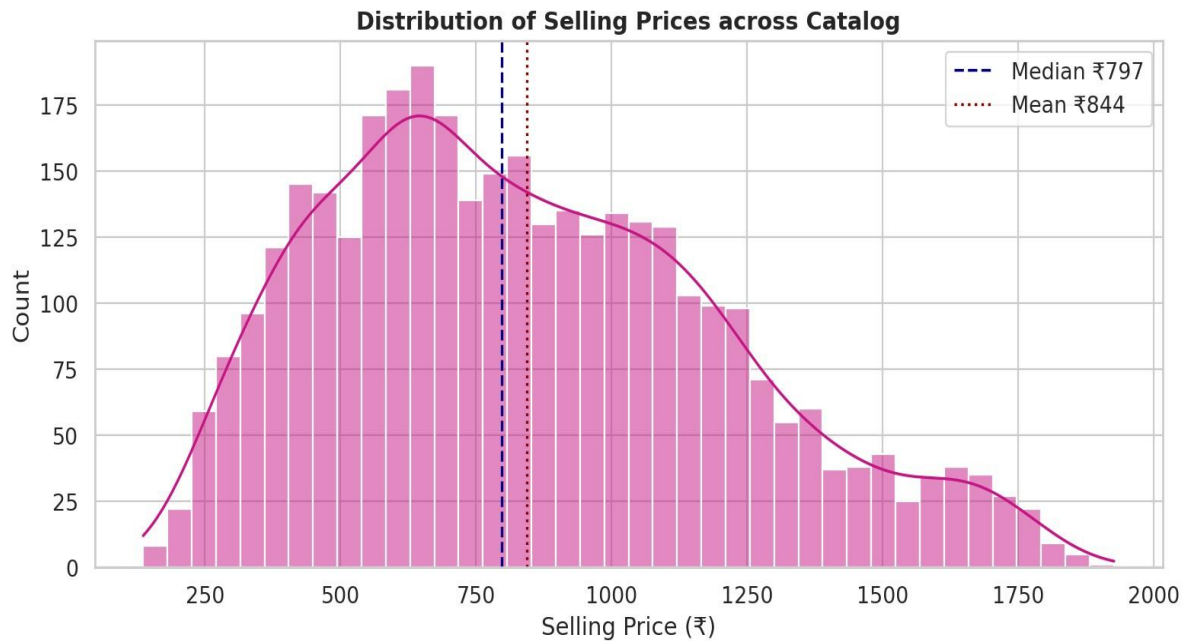


Figure 6. Distribution of selling prices

INSIGHTS:

- The distribution is right-skewed: the median (≈ ₹797) is lower than the mean (≈ ₹844), confirming the presence of a higher-priced tail.
- This pattern is typical of an accessible-premium beauty catalog where everyday colour cosmetics dominate bycount while a smaller set of complexion and tool products pull the mean upward.
- Maximum observed selling price in the dataset approaches ₹1,930.

KPI Snapshot:

- **Median selling price:** ≈ ₹797
- **Mean selling price:** ≈ ₹844
- **Maximum selling price:** ≈ ₹1,927

9. MRP vs Selling Price Relationship

The scatter plot below relates each product's MRP to its discounted selling price, coloured by discount percentage. Points that fall further below the diagonal represent deeper discounts relative to MRP.



Figure 7. MRP vs selling price, coloured by discount %

INSIGHTS:

- Selling price rises broadly in line with MRP, as expected, but the colour gradient shows that discount depth is not uniform at a given price point.
- Deeper discounts appear across both mid and higher MRP bands, indicating that promotions are used for assortment management as well as entry-level value offers.
- This view helps category managers identify where promotional intensity is concentrated across the price spectrum.

10. Top Highest-Discount Products

The table below lists products with the deepest discounts in the supplied catalog. Deep discounts often signal either clearance of older shade runs, campaign hero offers, or inventory-rebalancing actions.

Product	Category	MRP (₹)	SP (₹)	Disc %
Lip Liner Shade 17	Lips	520	235	54.7%
Nail Lacquer Shade 5	Nails	380	172	54.7%
Brow Studio Pencil Shade 2	Brows	890	403	54.7%
Mirror Compact	Tools & Accessories	700	318	54.5%
Liquid Lipstick Shade 21	Lips	830	379	54.3%
Kohl Star 24HR Kajal Shade 3	Eyes	350	160	54.3%
Hydra Cloud Cushion Shade 21	Face	1830	836	54.3%
EyeCanvas Palette Shade 5	Eyes	800	366	54.2%

Hydra Cloud Cushion Shade 20	Face	1470	672	54.2%
Nail Strengthener Shade 3	Nails	470	215	54.2%

INSIGHTS:

- Deep-discount products ($\approx 50\%+$) appear across multiple categories rather than being confined to a single segment.
- High-discount items can temporarily boost daily purchase volume but should be monitored for margin impact and brand perception.
- Clearance-style discounts on older shades free up inventory capacity for new launches — a common practice in colour cosmetics.

11. Product Net Content Distribution

The chart below shows how product net content (grams or millilitres) is distributed across the catalog. Most colour-cosmetic items are compact (under 30 g/ml), consistent with lipsticks, kajals, blushes and single eyeshadows typical of quick-commerce and e-commerce beauty purchases.

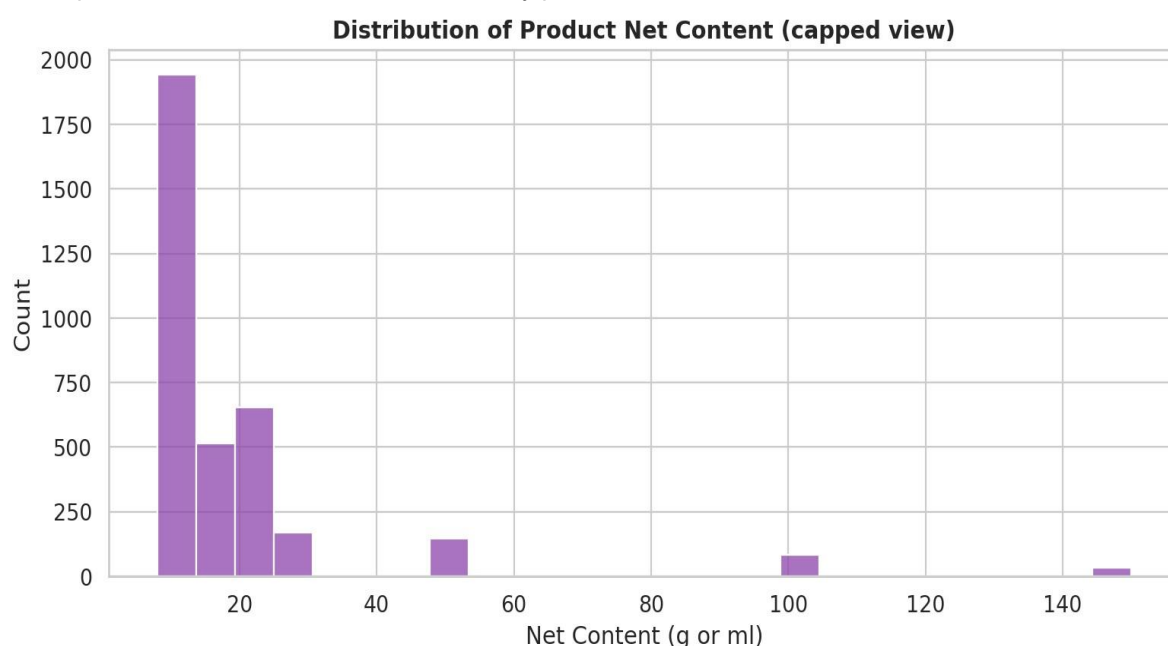


Figure 8. Distribution of product net content

INSIGHTS:

- Median net content is approximately **12 g/ml**; mean is higher because of a small number of larger tools and sets.
- Compact sizes support lower shipping cost and higher units-per-order in pure-play e-commerce channels.
- Weight/volume data can support logistics planning, packaging-cost estimation and sustainability reporting.

KPI Snapshot:

- **Median product net content: 12 g / ml**
- **Mean product net content: higher due to tail of larger items**

12. Average Daily Purchase Volume by Category

A core addition beyond pure catalog description is the daily purchase behaviour layer. The chart below shows average daily units purchased by category. Lips and Eyes lead in purchase velocity, reflecting both assortment depth and strong consumer demand for colour products.

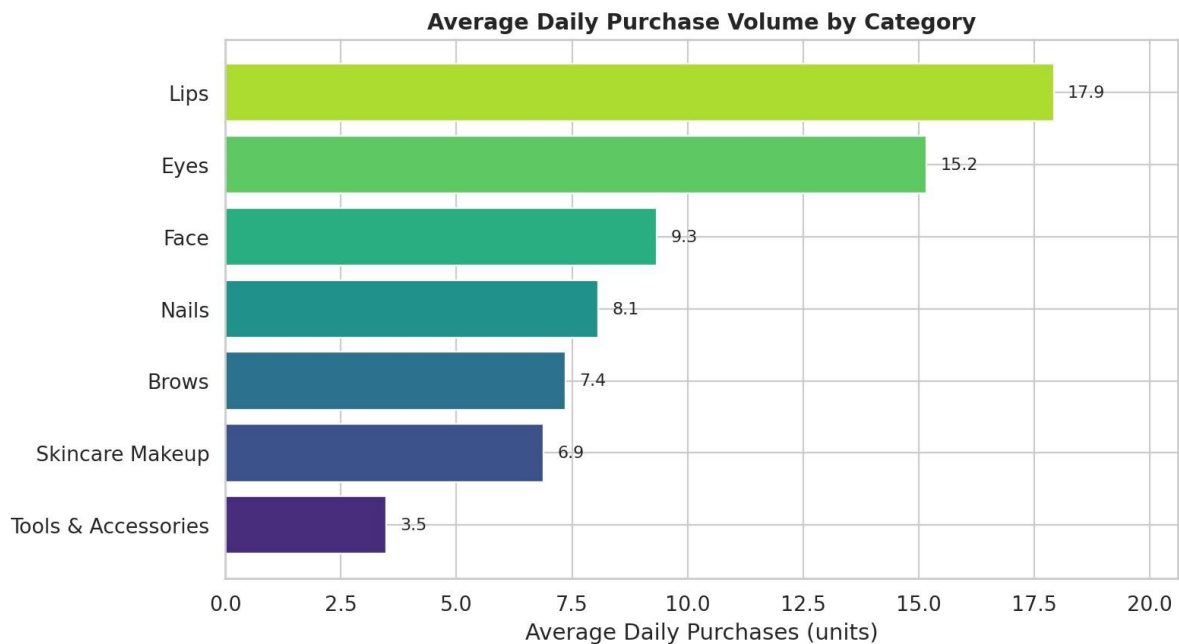


Figure 9. Average daily purchase volume by category

INSIGHTS:

- Lips exhibit the highest average daily purchase volume, followed by Eyes and Face.
- Tools & Accessories show the lowest average daily velocity, which is expected for higher-ticket, less frequent purchases.
- High-velocity categories should receive priority in inventory planning, warehouse slotting and campaign allocation.
- Mean catalog-wide average daily purchases per SKU is approximately **11.2 units**.

13. Category Contribution to 30-Day Unit Sales

Aggregating the last-30-day units sold reveals which categories contribute most to total volume. This is a critical view for assortment planning and for allocating marketing spend.

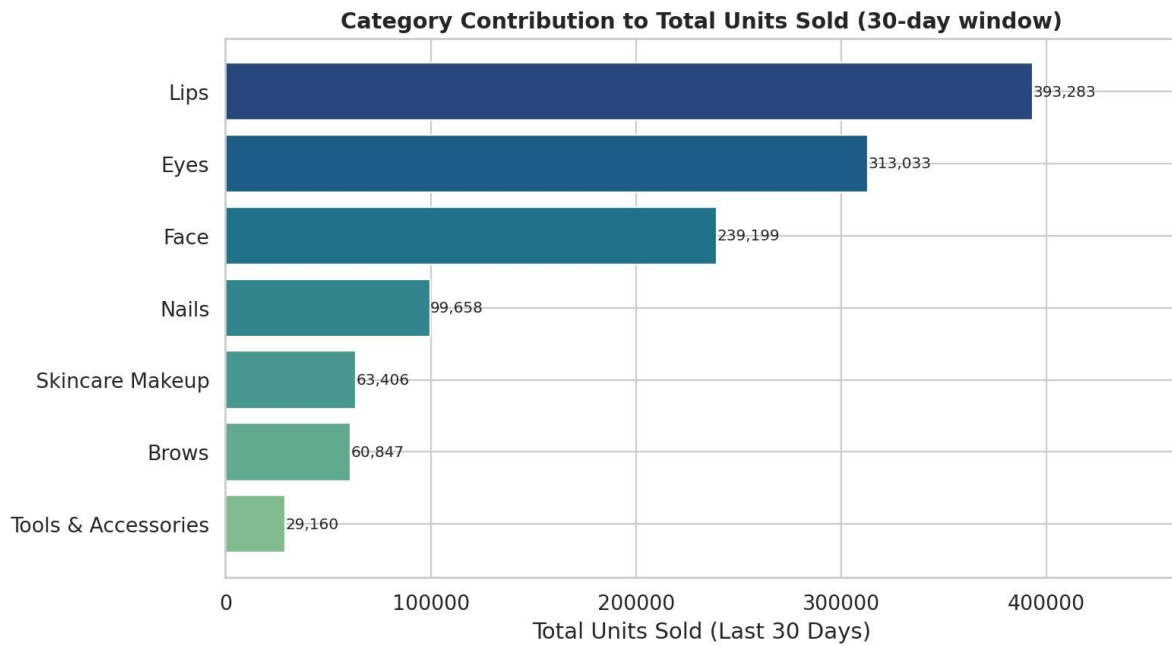


Figure 10. Total units sold in the last 30 days by category

INSIGHTS:

- Lips and Eyes together account for a dominant share of total unit volume in the 30-day window.
- Face, despite having the largest SKU count, contributes solidly but does not lead volume — indicating manyshade variants with more moderate individual velocity.
- Total units recorded across the catalog in the 30-day window exceed **1.19 million**.
- Volume concentration in a few categories implies that demand forecasting and replenishment efforts should be prioritised accordingly.

14. Price Sensitivity: Selling Price vs Daily Volume

The scatter below samples the relationship between selling price and average daily purchases, coloured by category. Lower-priced items generally show higher daily velocity, although category effects and brand affinity create substantial variation around the trend.



Figure 11. Selling price versus average daily purchases (sample)

INSIGHTS:

- A clear inverse relationship is visible: lower price points tend to attract higher daily purchase counts.
- Category still matters — some mid-priced Face and Lips SKUs sustain healthy volume because of brandstrength and shade desirability.
- Price-sensitivity analysis can guide the design of entry-price hero SKUs versus margin-accretive heroproducts.

15. Discount Depth vs Daily Purchase Volume

Deeper discounts are expected to lift short-term volume. The regression plot below explores this relationship across a sample of the catalog.

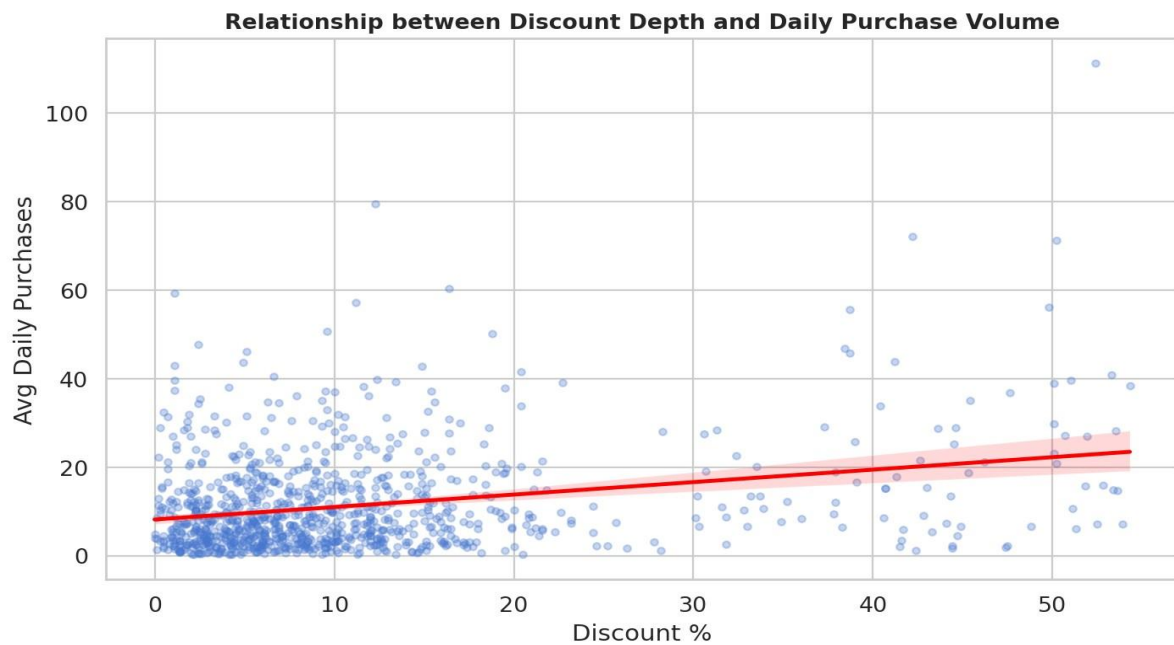


Figure 12. Relationship between discount percentage and daily purchases

INSIGHTS:

- There is a positive association between discount depth and daily purchase volume, as expected.
- The relationship is noisy: many high-volume SKUs sell well even at modest discounts, reflecting organic demand and brand pull.
- Heavy discounting should be used selectively (clearance, new-customer acquisition, festival peaks) rather than as a permanent lever.

16. Customer Rating Distribution

Customer ratings provide a quality and satisfaction signal that complements commercial metrics. The distribution below is tightly concentrated in the 4.0–4.8 range, consistent with a well-received accessible-premium brand.

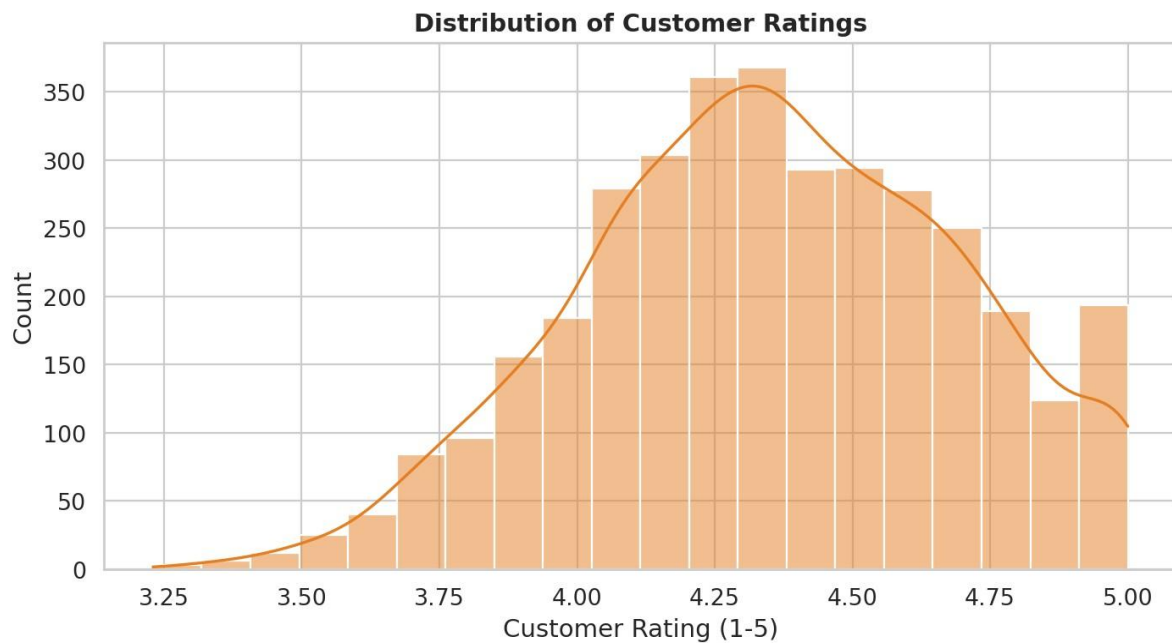


Figure 13. Distribution of customer ratings

INSIGHTS:

- Mean customer rating across the catalog is approximately **4.35 / 5**.
- The distribution is left-skewed toward high scores, indicating strong overall satisfaction.
- Products with ratings below 3.8 should be reviewed for formulation, shade accuracy or packaging issues.
- High ratings combined with high volume identify true hero products that deserve preferential inventory and marketing support.

17. Within-Category Price Dispersion

The boxplot visualises the spread of selling prices inside each category. Wide inter-quartile ranges indicate diverse price architecture (entry, mid, premium) within the same category; tight boxes indicate more uniform pricing.



Figure 14. Selling-price distribution within each category (boxplot)

INSIGHTS:

- Face and Tools show the widest price dispersion, reflecting both basic and premium offerings.
- Nails exhibits a tighter, lower price band — useful for frequent repurchase and gift/impulse occasions.
- Understanding within-category dispersion helps avoid cannibalisation between near-identical price points.

18. Top Products by Daily Purchase Volume

The following products currently lead the catalog in average daily units purchased. These are the commercial engines of the assortment and deserve priority in stock, content and paid media.

Product	Category	SP (₹)	Daily Units	30d Units	Disc %
Hydra Creme Lipstick Shade 12	Lips	614	111.3	3,156	52.4%
Lip Tint Shade 14	Lips	473	79.4	2,307	12.3%
Lip Liner Shade 14	Lips	398	72.1	1,694	42.2%
Lip Oil Shade 8	Lips	408	71.2	2,671	50.2%
Multi-Use Blush Shade 12	Face	568	70.2	1,920	41.4%
Kohl Star 24HR Kajal Shade 20	Eyes	291	69.4	1,544	19.0%
Lip Tint Shade 20	Lips	896	61.0	2,326	5.6%
Lip Oil Shade 23	Lips	618	60.3	2,217	16.4%
Primer Shade 24	Face	468	60.3	1,729	53.7%

Lip Liner Shade 3	Lips	567	60.0	1,760	11.4%
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Lips products dominate the top of the daily-volume ranking, reinforcing the category's role as the primary volume driver for the brand.

19. Category-Level Summary Metrics

The consolidated table below brings together the key descriptive KPIs for every category, enabling side-by-side comparison of assortment size, pricing, discount intensity, stock health, purchase velocity and customer sentiment.

Category	SKUs	Avg SP	Avg Disc	OOS %	Avg Daily	30d Units	Rating
Brows	280	■574	10.5%	15.0%	7.4	60,847	4.36
Eyes	680	■804	11.2%	13.1%	15.2	313,033	4.35
Face	850	■1155	11.4%	6.7%	9.3	239,199	4.37
Lips	720	■788	11.1%	9.9%	17.9	393,283	4.35
Nails	420	■413	9.7%	17.6%	8.1	99,658	4.33
Skincare Makeup	310	■880	9.3%	13.9%	6.9	63,406	4.33
Tools & Accessories	280	■1024	11.3%	11.4%	3.5	29,160	4.35

This single view is often the starting point for weekly category reviews. It surfaces outliers (high OOS, low rating, unexpectedly low velocity relative to SKU count) that then warrant deeper investigation at the SKU or shade level.

20. Overall Catalog-Health & Purchase-Behaviour Summary

The table below consolidates the headline metrics that describe the current state of the Kay Beauty catalog and its daily purchase dynamics.

Metric	Value
Total products (SKUs)	3,540
Categories tracked	7
Overall avg. selling price	≈ ■844
Overall avg. MRP	≈ ■949
Overall avg. discount	≈ 10.9%
Total estimated savings vs. MRP (1× all SKUs)	≈ ■3,71,957
Out-of-stock products	408 (≈ 11.5%)
In-stock products	3,132 (≈ 88.5%)
Median selling price	≈ ■797
Median product net content	12 g / ml
Mean avg. daily purchases (per SKU)	≈ 11.2 units
Total units sold (last 30 days)	≈ 1,198,586
Average customer rating	≈ 4.35 / 5

Taken together, the metrics portray a healthy, accessible-premium beauty catalog with strong purchase velocity in Lips and Eyes, solid average ratings, moderate discounting, and an out-of-stock rate that is manageable but not negligible. The Face category carries the largest assortment and the highest average price point, while volume leadership remains with colour products that consumers repurchase more frequently.

viral shades can convert potential advocates into frustrated customers. Implementing shade-level safety-stock rules for the top 15–20% of velocity SKUs would be a high-ROI operational improvement.

21. Next Steps: Moving from Descriptive → Predictive & Prescriptive

This report answers *what* the supplied catalog and recent purchase behaviour look like. The next analytical layers would combine richer order-level and customer-level data to explain *why* patterns occur and to forecast *what is likely to happen next*.

- **Demand / order forecasting by category and hero SKU** once daily or weekly sales time-series of sufficient length are available.
- **Stock-out prediction** using historical availability patterns, lead times and demand volatility to flag at-risk SKUs before they go to zero.
- **Dynamic pricing & promotion effectiveness analysis** linking discount depth, campaign calendar and sell-through to measure incremental volume versus margin trade-off.
- **Assortment optimisation** using SKU-level contribution margins, velocity and substitutability to recommend additions, delists or shade rationalisation.
- **Customer cohort & repurchase analysis** to understand how often buyers of Lips return for Face or Eyes, supporting cross-category journey design.
- **Sentiment & review-text mining** on low-rated SKUs to surface formulation or shade-match issues faster than aggregate star ratings alone.
- **International catalog comparison** (India vs UAE vs UK Space NK edit) to identify which SKUs travel well and which need local adaptation.
- **Trend and seasonality detection** when longer historical catalog and sales snapshots become available (festival peaks, wedding season, summer vs winter shade preferences).

In short: the descriptive foundation is now in place. Layering transactional, customer and time-series data on top of it will unlock the predictive and prescriptive capabilities that modern beauty brands use to stay ahead of both inventory risk and shifting consumer preference.

22. Glossary

MRP — Maximum Retail Price — the manufacturer-set ceiling price of a product before any discount is applied.

SKU — Stock Keeping Unit — a distinct, individually-tracked product/variant (including shade) in the catalog.

Discount % — The percentage reduction from MRP to arrive at the discounted selling price shown to the customer.

Out of Stock (OOS) — A product status flag indicating that available quantity is currently zero or unavailable for purchase.

Available Quantity — The recorded stock count for a product at the time the catalog snapshot was taken.

Category — A grouping of products by type (Lips, Eyes, Face, etc.) used for catalog organisation and reporting.

Avg. Daily Purchases — Estimated average number of units of a given SKU purchased per day over the observation window.

Units Sold (30 Days) — Total units of a SKU recorded as sold in the most recent 30-day period.

Net Content — The packaged quantity of product (grams or millilitres) declared on the pack.

Customer Rating — Average star rating (1–5 scale) submitted by customers who reviewed the product.

Accessible-Premium — Positioning that offers high-performance formulas and inclusive shade ranges at price points below traditional luxury while remaining above mass-market drugstore levels.

Hero SKU / Hero Shade — A high-velocity, high-visibility product or shade that disproportionately drives brand awareness and volume.

This report answers what the supplied catalog and daily purchase behaviour look like today. The next analytical layer would combine this catalog data with order/transaction-level and customer-level data to explain why stock or pricing patterns occur, and to forecast what is likely to happen next.

Prepared by **Laher Kashyap** (Roll No. 1250258252)
BCA (Data Science & Artificial Intelligence) | Section BCADS24
13th August 2026